

Technology Stack Template

Date	15 March 2026
Team ID	Batch 2023-27 ACOE
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component	Technology Chosen	Justification / Purpose
1	Machine Learning	XGBoost 1.7 + Scikit-learn	Best accuracy (96.55%) among 4 tested algorithms; fast inference
2	Data Processing	Pandas + NumPy	Efficient tabular data manipulation and array operations
3	Model Serialization	Joblib	Save and load XGBoost model and StandardScaler
4	Backend / Server-Side	Python 3.11 + Flask	Lightweight WSGI framework; integrates directly with ML objects
5	Frontend / Client-Side	HTML5 + CSS3 + JS	Responsive design; no framework overhead
6	Templating Engine	Jinja2	Server-side HTML rendering; template inheritance
7	Cloud / Hosting	IBM Cloud Foundry	Free-tier cloud hosting; seamless deployment
8	Containerization	Docker	Reproducible deployment environment; easy scaling

S.No	Architecture Component	Technology Chosen	Justification / Purpose
9	Version Control	Git + GitHub	Source code management; collaborative development